PROTOTYPE A

RING PAN-TILT INDOOR CAMERA

FACTS ONLY - No privacy score or rating

PRIVACY INFORMATION

Who Can Access?	You and the Shared Users you authorise can access the camera.
Video Storage	Recorded videos are stored in the Ring cloud when recording features are used with an eligible Ring subscription.
Manufacturer Access	Optional End-to-End Encryption (E2EE) can be enabled. When E2EE is enabled, Ring services cannot access your encrypted video content.
Data Sharing	Some personal information may be processed by Ring affiliates and service providers for defined purposes such as storage, customer service and analytics.
Video Retention	For event videos, you can choose a storage period from **1 to 180 days.** The default is **30 days**.
Your Controls	Manage Shared Users; delete recordings; change event-video retention time; enable optional E2EE.
Physical Privacy	Closing the physical Privacy Cover turns off the camera and microphone.

Research question: Can users understand the privacy information from facts alone?

PROTOTYPE B

RING PAN-TILT INDOOR CAMERA

RATING LABEL - Same product information, summarised with two research ratings

DATA ACCESS EXPOSURE	USER CONTROL
HIGH	**HIGH**
6 / 10	**7 / 10**
Higher exposure means more CCTV data may leave the local environment or involve external services.	Higher control means the user has more options to manage access, storage, deletion, retention and data use.

KEY PRIVACY FACTS

Who Can Access?	**You and authorised Shared Users.**
Video Storage	Cloud-based recording with an eligible Ring subscription.
Manufacturer Access	Depends on encryption settings. Optional E2EE prevents Ring services from accessing encrypted video content.
Video Retention	Event videos: 1-180 days; default 30 days.
Physical Privacy	Privacy Cover turns off the camera and microphone.

WHY THIS RATING?

Data Access Exposure	Score	User Control	Score
Manufacturer Access	1/2	Access Control	2/2
Other User Access	1/2	Storage Control	0/2
Cloud Exposure	2/2	Deletion Control	2/2
Third-party Sharing	1/2	Retention Control	2/2
Secondary Data Use	1/2	Consent/Data-use Control	1/2
Total	**6/10**	**Total**	**7/10**

Prototype research ratings based on publicly available official Ring information. These are not official Ring privacy ratings.

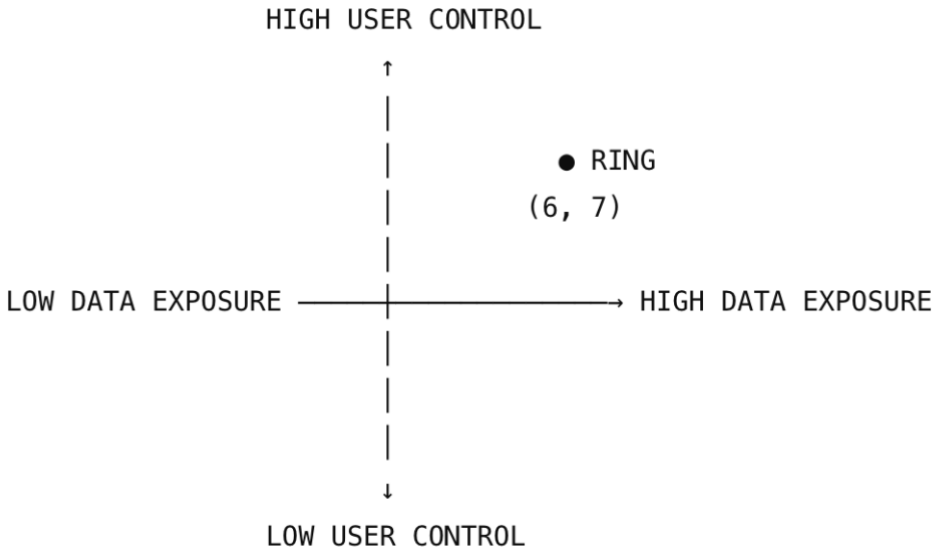
PROTOTYPE C

RING PAN-TILT INDOOR CAMERA

XY PRIVACY PROFILE - Same product information shown as a two-dimensional profile

PRIVACY PROFILE

Data Access Exposure: 6 / 10 User Control: 7 / 10



HOW TO READ THIS PROFILE

Data Access Exposure (left to right): Further right means more CCTV data may leave the local environment or involve external services.

User Control (bottom to top): Higher means the user has more options to manage access, storage, deletion, retention and data use.

WHY IS IT HERE?

Exposure 6/10	**Manufacturer Access 1/2; Other User Access 1/2; Cloud Exposure 2/2; Third-party Sharing 1/2; Secondary Data Use 1/2.**
Control 7/10	Access Control 2/2; Storage Control 0/2; Deletion Control 2/2; Retention Control 2/2; Consent/Data-use Control 1/2.
Key Facts	Cloud-based recording; event-video retention 1-180 days (default 30); authorised Shared Users; optional E2EE; physical Privacy Cover.

Prototype research profile based on publicly available official Ring information. This is not an official Ring privacy rating.

1. Testing Introduction

We will show you three different privacy information designs for the same CCTV camera.
Please think aloud while looking at each design. Tell us what you are looking at, what you think it means, and anything you find confusing.
We are testing the design, not you.

2. **Part 2 — Think Aloud Instructions**

Think Aloud

While looking at each prototype, please think aloud.

This means I would like you to say what you are thinking while you look at the information.

For example, you can tell me:

“I am looking at this section first.”

“I think this means…”

“I don't understand this.”

“I am looking for information about…”

“This number is confusing.”

“I think this camera has…”

You don't need to explain everything perfectly. Just say what naturally comes to your mind.

What are you thinking now?

Can you tell me what you are looking at now?

Please keep thinking aloud.

3. **Q1 — Manufacturer Access**

Can Ring access your encrypted recordings? Under what condition?

Q2 — User Control

What privacy controls do you have?

Q3 — Storage

Where are your recordings stored?

Q4 — Confidence

How confident are you that you understand this privacy information? (1–5)

Time: ____ seconds
Accuracy: ____ / 3
Confidence: ____ / 5

4. A/B/C

Q1：Which prototype was easiest to understand? Why?

Q2：Which prototype helped you find information fastest?

Q3：Which prototype do you prefer? Why?

Q4：Was anything confusing or difficult to understand?

Q5：Was any important privacy information missing?

1 = like
2 = sec
3 = dislike

Prototype	Rank
A — Facts Only	___
B — Rating Label	___
C — XY Privacy Profile	___

Easiest: A / B / C
Fastest: A / B / C
Preferred: A / B / C
Why: _____

Measure	Prototype A	Prototype B	Prototype C
Q1 Manufacturer Access	Correct / Partial / Wrong	Correct / Partial / Wrong	Correct / Partial / Wrong
Q2 User Control	Correct / Partial / Wrong	Correct / Partial / Wrong	Correct / Partial / Wrong
Q3 Storage	Correct / Wrong	Correct / Wrong	Correct / Wrong
Overall Accuracy	___ / 3	___ / 3	___ / 3
Completion Time	___ sec	___ sec	___ sec
Confidence	___ / 5	___ / 5	___ / 5
Observed Confusion			
Important Quote			
Preference Rank	___	___	___

Easiest to understand:
A / B / C

Fastest to find information:
A / B / C

Most preferred:
A / B / C

Main confusion:

Missing information requested by participant:

Important Think Aloud quote:

“_____”

Researcher notes:

Participant Testing Order

P1 A → B → C

P2 B → C → A

P3 C → A → B

P4 A → C → B

P5 B → A → C

P6 C → B → A